

DL & GenAI Project — Smart MCQ Solver

Milestone 3: Context Augmentation with RAG Pipeline

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Overview

Milestone 3 explores Retrieval-Augmented Generation (RAG) as a strategy to improve MCQ prediction by supplying the model with relevant external context before it makes a prediction. A standalone model must answer questions purely from patterns learned during training. RAG addresses this limitation by first retrieving semantically similar documents from a knowledge base and prepending that context to the input, giving the model additional grounding to reason from.

In this milestone, the knowledge base is constructed entirely from the training set — specifically, the correct answer text for each training row. A FAISS vector index is built over these entries using MiniLM embeddings. At inference time, a query prompt is encoded, the top-k most similar entries are retrieved, a cross-encoder reranks them, and the best document is prepended as context before passing the augmented input to a zero-shot classifier. The end-to-end pipeline is evaluated on a 20-row sample with mAP@3 as the metric.

Pipeline Architecture

The RAG pipeline consists of four sequential components:

- 1. Knowledge Base Construction.** The training set is used to build the knowledge base. For each row, the text of the correct answer option is extracted and stored as a document. This gives 2000 documents — one per training question — representing verified correct answers.
- 2. Dense Retrieval with FAISS.** All 2000 documents are encoded using `all-MiniLM-L6-v2` (a sentence transformer) into 384-dimensional vectors. These are indexed using a FAISS `IndexFlatL2` index for exact L2 nearest-neighbour search. At query time, the prompt is encoded with the same model and the top-k most similar documents are retrieved.
- 3. Cross-Encoder Reranking.** The top-10 retrieved documents are reranked using the cross-encoder model `cross-encoder/ms-marco-MiniLM-L-6-v2`. Unlike the bi-encoder used in retrieval, a cross-encoder scores each (query, document) pair jointly, giving a more accurate relevance signal. The highest-scoring document after reranking is selected as the context.
- 4. Zero-Shot Classification with Augmented Input.** The augmented input is constructed as:

Context: {best_doc} Question: {prompt}

This string is passed to `facebook/bart-large-mnli` via HuggingFace's zero-shot classification pipeline. The five answer options are provided as candidate labels and the model returns a probability score for each, from which top-3 predictions are derived.

Implementation Details

Libraries Used

`faiss-cpu` for vector indexing, `sentence-transformers` for bi-encoder embeddings and cross-encoder reranking, `transformers` for the zero-shot classification pipeline, and `sklearn` for TF-IDF (available but superseded by the dense retrieval approach).

Knowledge Base Design

Only correct answer texts are stored in the knowledge base — not full question-option combinations. This is a deliberate design choice: storing only correct answers means retrieved documents are always semantically meaningful answers, reducing noise during context injection.

Retrieval Parameters

Top-10 documents are retrieved in the first stage (bi-encoder), which are then reranked by the cross-encoder to select the single best document. For the mAP@3 evaluation, top-5 retrieval with cross-encoder reranking was used to select the best context document per query.

Token Budget Analysis

For a sample query (row 42), concatenating the top-5 retrieved documents with the prompt produced a RAG string of significant length. Tokenizing with `bert-base-uncased` revealed the total token count, confirming the input fits within standard transformer context windows.

Adversarial Context Experiment

An explicit adversarial test was conducted: an irrelevant document (kb[999]) was injected as context for the question at row 150. This allowed a direct comparison of model behaviour across three conditions — no context, correct context, and adversarial context — to understand how sensitive the zero-shot model is to context quality.

Results and Evaluation

Single-Question Analysis (Row 150)

Condition	Probability of Correct Option
No RAG (baseline zero-shot)	0.384
RAG with correct context injected	0.989
RAG with adversarial context injected	(recorded in notebook)

The jump from 0.384 to 0.989 when the correct context is injected is a strong signal that the zero-shot model can leverage retrieved context effectively. The adversarial experiment demonstrated how incorrect context degrades performance, underlining that retrieval quality is the critical bottleneck in a RAG system.

Retrieval Quality

On a 100-row hit-rate evaluation, the FAISS bi-encoder retrieved the true document in the top-5 results with a hit rate recorded as Q7 in the notebook. The true document for row 150 was found at rank 1 in bi-encoder retrieval, and cross-encoder reranking maintained it at the top position.

End-to-End mAP@3

The full RAG pipeline (retrieve top-5, cross-encoder rerank, augment, zero-shot classify) was evaluated on 20 training rows:

Model	mAP@3
RAG Pipeline (20-row sample)	reported as Q8 in notebook
MiniLM baseline (Milestone 1)	0.3996
BiLSTM from scratch (Milestone 2)	0.9983

Observations and Insights

Context quality is everything. The single-question experiment showed that injecting the correct document nearly triples the model's confidence in the right answer (0.384 to 0.989). Conversely, adversarial context degrades performance, showing the model is highly sensitive to what is prepended.

The dataset's near-duplicate structure aids retrieval. As established in the EDA from the main notebook, 97% of test prompts have a near-identical counterpart in the training set. This means FAISS retrieval will almost always find the relevant document, making this dataset particularly well-suited for a RAG approach.

Cross-encoder reranking adds precision. The bi-encoder retrieves by approximate similarity in embedding space. The cross-encoder scores pairs jointly and produces a more calibrated relevance score, which is especially important when retrieved documents are semantically similar to each other.

Computational cost is a practical concern. Running `facebook/bart-large-mnli` for zero-shot classification on every query is significantly slower than the BiLSTM inference, which completes in milliseconds. The RAG pipeline as implemented is not suitable for real-time deployment without optimisation.

Conclusion

Milestone 3 demonstrated a complete RAG pipeline for MCQ answering, combining FAISS-based dense retrieval, cross-encoder reranking, and zero-shot classification with context augmentation. The pipeline showed strong single-question performance when retrieval is accurate, validating the core RAG hypothesis. The milestone contributes a reusable retrieval and augmentation framework to the overall project and provides a foundation for more sophisticated ensemble strategies in subsequent milestones.